

Proposed Solution

Date	26 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To develop a machine learning-based crop recommendation system that helps farmers choose the most suitable crop based on soil nutrients and environmental conditions, improving productivity and sustainable farming.
Scope	The system analyzes soil parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend the best crop using machine learning algorithms through a Flask web application.
Problem Statement	
Description	Farmers often face difficulty in selecting the right crop because they lack proper knowledge about soil nutrients and weather conditions. This may lead to poor yield and financial loss.
Impact	Providing accurate crop recommendations helps farmers increase productivity, reduce crop failure, improve resource utilization, and support sustainable agriculture.
Proposed Solution	
Approach	Develop a Flask-based web application integrated with machine learning models such as KNN, Logistic Regression, Decision Tree, Random Forest, and K-Means to analyze agricultural data and recommend suitable crops.
Key Features	- Machine learning-based crop recommendation. - User-friendly web interface.

	• Real-time crop prediction. • Accurate recommendations using soil and climate data.
--	---

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i3 or above
Memory	RAM	Minimum 4 GB
Storage	Disk Space	Minimum 10 GB Free Space
Software		
Frameworks	Web Framework	Flask
Libraries	Python Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Agricultural Dataset	Crop_recommendation.csv (Kaggle Dataset)